Assignment 01

PUBH 8878

1. Make a Zotero account using the guide here. Make sure you use your GW email address, as this will provide unlimited cloud storage for PDFs. Once you have created your account, email chiraaggohel@gwu.edu your username.
2. (Laird, 2.4) How many genotypes are possible with a 3-allele marker? With K alleles?
3. (Laird, 2.6) Consider a recessive Mendelian disease, where in the population, $P(\text{an individual has 2 disease variants}) = 0.000001$.

    a. What is the probability that a randomly selected person is affected? Suppose that the randomly selected person is affected. What does that imply about the probability that their sibling is also affected (you can assume that having either one or two parents with two variants is so rare that you can ignore them)?

    b. Now answer both of these questions assuming the penetrance is only $\frac{1}{2}$, i.e., $P(\text{disease}|2 \text{ variants}) = \frac{1}{2}$, but the phenocopy rate is still zero.

4. Consider a sample size of n of unrelated haploid individuals is obtained from some population with the objective of estimating allele frequency at a biallelic locus. The sample contains x copies of A, and $n - x$ copies of a.

    a. Plot the probability distribution of X given $n = 30$, and $\theta = .1$. Plot the probability distribution of X given $n = 1000$, and $\theta = .1$.
    b. Lets say we observed 30 samples, with 10 copies of allele A. Plot the likelihood function for θ
    c. What is the MLE of θ?
    d. Let's say $n = 1000$, and $x = 100$. What is the sampling variance of $\hat{\theta}$?
    e. Let's say $n = 100$, and $x = 10$. What is the sampling variance of $\hat{\theta}$? Why is this different than the result above?

5. Refer to equations (1.3) and (1.5) in Sorensen. Say you observe 8 individuals, and 1 copies of genotype X. Assume that $X \sim \mathsf{Binom}(n, \theta)$.

a. Compute $\hat{\theta}$

b. Compute $\widehat{\text{Var}}(\hat{\theta})$
c. Provide a 95% Wald confidence interval for θ
d. Write an interpretation of this confidence interval. What problem does this reveal about the Wald confidence interval?
e. Compute a 95% Wilson confidence interval for θ. Documentation for this can be found here. *Hint: you will need the* `fastR2` *package.*

6. Refer to slides 13 and 14 in lecture 1. Write a one to two sentence answer for how a researcher would try to answer each question.

7. What is your math background? What is your programming background?